

Mapping the Group–Individual Fairness Trade-off: An Empirical Study with a Differentiable Composite Loss

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Abstract—Algorithmic fairness is not a single objective. Group fairness asks that demographic groups receive comparable treatment on average, while individual fairness asks that the burden of a model’s errors be distributed evenly across the people subject to it. Theoretical results establish that such criteria are frequently incompatible, but the magnitude of the resulting trade-off on real data is far less documented. We introduce a differentiable composite loss whose single parameter β interpolates continuously between a soft demographic-parity term and a soft generalized-entropy term, and we use it as an instrument to map the trade-off empirically. Sweeping a 7×7 grid over the accuracy–fairness weight α and the group–individual weight β , across three standard datasets (COMPAS, German Credit, Adult), two model families (logistic regression and a multilayer perceptron), and ten random seeds, we train and evaluate 2,940 models. We observe a statistically significant double dissociation: optimizing the group term reduces the demographic parity difference by up to 89% but significantly worsens individual-fairness inequality and, on Adult, degrades equalized odds by a factor of 3.3; optimizing the individual term significantly improves individual fairness while leaving group disparity unchanged or slightly worse. The composite loss attains group-fairness levels comparable to established pre-, in-, and post-processing baselines. We further report a negative result with a concrete diagnosis: on German Credit no effect survives significance testing, and we show this is a failure of validation-based model selection at $n = 1000$ —validation and test demographic parity are essentially uncorrelated ($r = 0.08$, versus 0.94 and 0.99 on the other two datasets)—rather than a failure of the method. Our results quantify a trade-off that practitioners must resolve by policy, not by tuning.

Index Terms—algorithmic fairness, group fairness, individual fairness, demographic parity, generalized entropy index, fairness–accuracy trade-off, differentiable surrogate loss

I. Introduction

Machine-learned models increasingly mediate consequential decisions about people: whether a defendant is labelled high-risk, whether an applicant receives credit, whether a résumé is advanced. A decade of work has established that such models can encode and amplify disparities present in their training data [1]–[3]. What is far less settled is what should be done about it—because “fairness” is not one property but a family of mutually inconsistent ones.

Two families dominate practice. Group fairness requires that outcomes be comparable across demographic groups; its canonical instance, demographic parity, requires equal selection rates. Individual fairness requires that similar individuals be treated similarly [4], and in its influential outcome-based operationalization requires that the benefit each person derives from the classifier be distributed evenly across the population [5]. These are different requirements, and satisfying one does not imply satisfying the other.

That they can conflict is known theoretically. Calibration and error-rate balance cannot generally hold at once when base rates differ [6], [7]; demographic parity and equalized odds are likewise incompatible under the same condition [8]. What these results do not supply is a magnitude. A practitioner told that “you cannot have both” still needs to know what one costs in units of the other, on data resembling their own, with uncertainty quantified. Comparative empirical studies of fairness interventions exist [9], but they typically compare discrete methods, each committed in advance to a single fairness notion. What is missing is a study that varies the fairness objective continuously and measures both families of metrics along the way.

A. Contributions

We build such an instrument and use it to map the trade-off.

- 1) A differentiable composite loss (Section III) whose parameter β interpolates continuously between a soft demographic-parity term and a soft generalized-entropy term, with a second parameter α controlling the overall weight on fairness versus accuracy. Both fairness terms are computed on raw predicted probabilities so that gradients flow through them—the thresholding step that defines the evaluation metrics is not differentiable and cannot be optimized directly.
- 2) An empirical trade-off map (Section V) built from a 7×7 (α, β) grid across three standard datasets, two model families, and ten seeds: 2,940 trained models,

all metrics reported as mean $\pm$ standard deviation with paired significance tests.

- 3) A statistically significant double dissociation. Optimizing the group term reduces the demographic parity difference by 78%–89% ($p_{\text{Holm}} = 0.008$) while significantly worsening individual-fairness inequality; optimizing the individual term significantly improves individual fairness while leaving group disparity unchanged or significantly worse. On Adult, driving demographic parity to near zero degrades equalized odds by a factor of 3.3, an empirical instance of a known impossibility.
- 4) A diagnosed negative result. On German Credit no effect survives correction. We show this is not a failure of the method but of validation-based model selection at small n : validation and test demographic parity are effectively uncorrelated ($r = 0.08$ versus 0.94 and 0.99 elsewhere), so the selection rule cannot identify fair configurations. Under a pre-specified configuration the effect is present. We believe this sample-size floor on fairness model selection is of independent practical interest.
- 5) A benchmark soundness correction. We identify and remove a label-leaking feature from a COMPAS feature set in prior use, and report the honest accuracy that results.

The composite loss is the instrument; the trade-off map is the contribution. We do not claim a new state of the art in fairness mitigation, and Section V shows the loss matches rather than beats established baselines on group-fairness metrics. Its value is that, unlike those baselines, it exposes a continuous dial between two fairness notions, which is precisely what mapping a trade-off requires.

II. Related Work

A. Group Fairness Criteria and Mitigation

Group-fairness criteria constrain statistics of the prediction conditioned on a sensitive attribute. Demographic parity equalizes selection rates; equalized odds equalizes true- and false-positive rates across groups [10]; equal opportunity equalizes true positive rates only. Surveys catalogue more than twenty such definitions [3], [11], many of which cannot hold simultaneously.

Mitigation methods are conventionally grouped into three families, all three of which we use as baselines. Pre-processing alters the training data: reweighing assigns each sample a weight making label and sensitive attribute independent in the weighted distribution [12]; related work repairs feature distributions directly [13] or learns fair representations [14]. In-processing alters the objective, either by adding fairness constraints or penalties to the training problem [15], by reduction to a sequence of cost-sensitive problems [16], or adversarially [17]. Post-processing adjusts a fixed model’s outputs, typically via group-dependent thresholds [10]. Our method is in-

processing; we compare against a representative of each family.

B. Individual Fairness

Dwork et al. [4] formalize individual fairness as a Lipschitz condition: individuals who are similar under a task-specific metric must receive similar output distributions. The framework is principled but requires a similarity metric that is rarely available in practice.

Speicher et al. [5] propose an outcome-based alternative that requires no similarity metric. For each individual they define a benefit b_i capturing how much that person received relative to what they were owed, and then apply an inequality index from the economics literature—the generalized entropy (GE) family [18], [19]—to the benefit vector. This unifies group and individual unfairness: the GE index of the benefit vector decomposes additively into a between-group and a within-group component. We adopt this operationalization, and we are explicit throughout that it measures inequality of outcomes rather than the Lipschitz condition of [4].

C. Impossibility and Trade-off Results

Kleinberg et al. [6] and Chouldechova [7] independently showed that calibration, balance for the positive class, and balance for the negative class cannot hold together unless base rates are equal or prediction is perfect. Corbett-Davies et al. [20] characterize the cost of imposing parity constraints on a utility-maximizing decision rule. Barocas et al. [8] give the general statement that independence (demographic parity) and separation (equalized odds) are incompatible whenever the sensitive attribute and label are dependent—the condition that holds in all three of our datasets.

These results establish existence, not magnitude, and they concern conflicts within group fairness. Our focus is the less-charted conflict between group and individual fairness, measured rather than derived.

D. Empirical Comparisons and Benchmark Critique

Friedler et al. [9] benchmark fairness-enhancing interventions across datasets and metrics, and emphasize that conclusions are sensitive to preprocessing and train/test splits—motivating our per-seed protocol. We extend this style of comparison along a new axis: rather than comparing methods that each fix a fairness notion, we vary the notion continuously within a single method.

A parallel literature critiques the benchmarks themselves. Bao et al. [21] argue that COMPAS is routinely used in ways that misrepresent the underlying criminal-justice process; Fabris et al. [22] survey the provenance of fairness datasets; Ding et al. [23] document defects in the standard Adult dataset and propose replacements. We contribute a concrete instance: Section IV identifies a label-leaking feature in a COMPAS feature set and quantifies the inflated accuracy it produces.

III. Method

A. Problem Setting and Notation

We consider binary classification. Let $x_i \in \mathbb{R}^d$ be the feature vector of individual i , $y_i \in \{0, 1\}$ the true label, and $s_i \in \mathcal{S}$ the sensitive attribute, where $\mathcal{S}$ is a finite set of demographic groups. A model f_θ produces a logit $z_i = f_\theta(x_i)$, a probability $p_i = \sigma(z_i)$ with $\sigma(\cdot)$ the logistic function, and a hard decision $\hat{y}_i = \mathbf{1}[p_i \geq 0.5]$.

B. Why a Surrogate Is Necessary

The evaluation criteria of Section IV are defined on the hard decisions $\hat{y}_i$. The map $p_i \mapsto \mathbf{1}[p_i \geq 0.5]$ has zero derivative almost everywhere and is undefined at the threshold, so any objective written in terms of $\hat{y}$ has no usable gradient and cannot be minimized by backpropagation. We therefore construct differentiable surrogates that operate on the probabilities p_i directly, and verify empirically (Section V-A) that each surrogate tracks the hard metric it stands in for.

C. Group Term: Soft Demographic Parity

The demographic parity difference is the largest gap in selection rate between any two groups. Replacing the selection rate with the mean predicted probability gives a differentiable analogue:

$$\mathcal{L}_{\text{DP}}(p, s) = \max_{g \in \mathcal{S}} \bar{p}_g - \min_{g \in \mathcal{S}} \bar{p}_g, \quad \bar{p}_g = \frac{1}{|\mathcal{I}_g|} \sum_{i \in \mathcal{I}_g} p_i, \quad (1)$$

where $\mathcal{I}_g = \{i : s_i = g\}$. Equation (1) is differentiable in θ through p , and converges to the hard demographic parity difference as the probabilities saturate towards $\{0, 1\}$.

D. Individual Term: Soft Generalized Entropy

Following Speicher et al. [5] we define the benefit received by individual i as

$$b_i = 1 + p_i - y_i. \quad (2)$$

A correctly served individual has $b_i \approx 1$; a false positive receives more than deserved ($b_i \rightarrow 2$) and a false negative less ($b_i \rightarrow 0$). Individual unfairness is then the inequality of the benefit vector, measured by the generalized entropy index of order γ :

$$\mathcal{L}_{\text{GE}}(p, y) = \frac{1}{n \gamma (\gamma - 1)} \sum_{i=1}^n \left[\left(\frac{b_i}{\bar{b}} \right)^\gamma - 1 \right], \quad (3)$$

where $\bar{b} = n^{-1} \sum_i b_i$ is the mean benefit. We use $\gamma = 2$ throughout, for which (3) equals half the squared coefficient of variation of b . The index is zero exactly when every individual receives identical benefit, and grows as the burden of the model's errors concentrates on fewer people. Because b_i depends on p_i smoothly, (3) is differentiable.

E. The Composite Loss

The two fairness terms are combined with the accuracy term into a single objective with two interpolation weights $\alpha, \beta \in [0, 1]$:

$$\mathcal{L}(\theta) = \alpha \mathcal{L}_{\text{BCE}} + (1 - \alpha) \left[\beta \mathcal{L}_{\text{DP}} + (1 - \beta) \mathcal{L}_{\text{GE}} \right] \quad (4)$$

where $\mathcal{L}_{\text{BCE}}$ is binary cross-entropy computed from logits for numerical stability. The two parameters have distinct roles:

- α trades accuracy against fairness. At $\alpha = 1$ the loss reduces to plain cross-entropy, giving the unconstrained baseline. As $\alpha \rightarrow 0$ accuracy is ignored entirely, which yields degenerate constant predictors (Section III-G).
- β trades group against individual fairness, and is the variable of interest in this study. At $\beta = 1$ the fairness budget is spent entirely on demographic parity; at $\beta = 0$ entirely on benefit equality; intermediate values blend them.

Sweeping β at fixed α therefore holds the total fairness pressure constant while redistributing it between the two notions—which is exactly the comparison needed to isolate a trade-off between them.

F. Models

We instantiate (4) with two model families, chosen to span a linear and a nonlinear hypothesis class while remaining standard in the fairness literature:

- Logistic regression: a single affine layer returning a logit.
- MLP (64–64): two hidden layers of 64 units with ReLU activations, returning a logit.

Both return logits rather than probabilities; the sigmoid is applied once, inside the loss.

G. Training and Selection Procedure

Algorithm 1 states the full procedure. Training uses Adam [24] with learning rate 0.01 and weight decay 10^{-4} , for at most 300 epochs, with early stopping on the validation value of the same composite loss being optimized (patience 20) and restoration of the best weights from a deep copy. Because (1) requires all groups to be present in order to form a group gap, each epoch is a single full-batch gradient step rather than a pass of mini-batches.

Two details in Algorithm 1 matter for the validity of the results. First, test metrics are never used for selection: the (α, β) pair is chosen by distance to the utopia point (accuracy = 1, $\text{DPD}_2 = 0$) computed on validation data alone. Second, the constraint $0.05 \leq \hat{r}_{\text{va}} \leq 0.95$ on the validation positive rate $\hat{r}_{\text{va}}$ excludes near-constant predictors: a model that assigns every individual to one class attains $\text{DPD}_2 = 0$ trivially while being useless. This guard is not cosmetic—at $\alpha = 0$ we observe validation positive rates above 0.99.

Algorithm 1 $\alpha \times \beta$ sweep with validation-only selection

Require: dataset D , seed set S , grids A (for α), B (for β)

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1: for each seed  $k \in S$  do
2:    $(D_{\text{tr}}, D_{\text{va}}, D_{\text{te}}) \leftarrow \text{Split}(D, 60/20/20, k)$ 
3:   fit scaler and encoder on  $D_{\text{tr}}$ ; apply to all splits
4:   for each architecture  $a$  and each  $(\alpha, \beta) \in A \times B$  do
5:     seed weights with  $k$ ; initialise  $f_\theta$ 
6:     train  $f_\theta$  on  $D_{\text{tr}}$  minimising (4)
7:     early-stop on  $\mathcal{L}$  evaluated on  $D_{\text{va}}$ 
8:     record test metrics, and validation accuracy,
       DPD2 and positive rate
9:   end for
10:  // selection uses validation quantities only
11:   $F \leftarrow \{(\alpha, \beta) : \alpha < 1, 0.05 \leq \hat{r}_{\text{va}} \leq 0.95\}$ 
12:   $(\alpha^*, \beta^*) \leftarrow \arg \min_F \sqrt{(1 - \text{acc}_{\text{va}})^2 + \text{DPD}_{2,\text{va}}^2}$ 
13:  report test metrics at  $(\alpha^*, \beta^*)$ 
14: end for
15: aggregate over  $S$ ; paired Wilcoxon test with Holm
   correction

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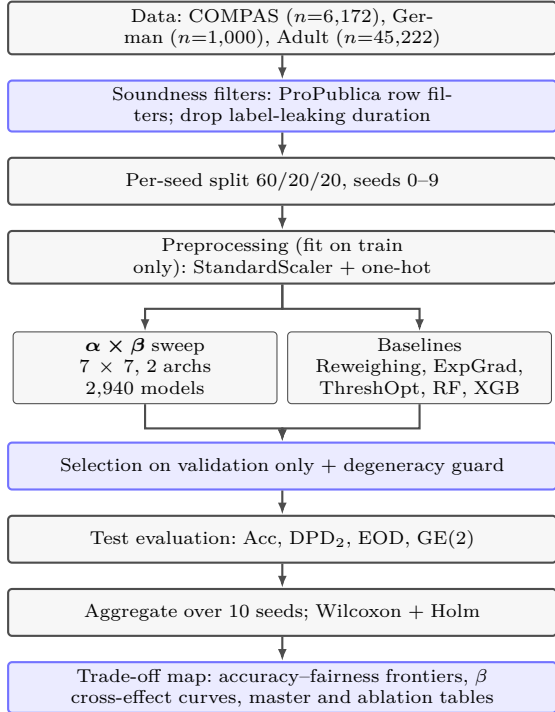


Fig. 1. Experimental pipeline. Shaded boxes mark the three steps that most affect the validity of the conclusions: removal of the label-leaking COMPAS feature, selection performed strictly on validation data with a degenerate-predictor guard, and reporting of the resulting trade-off map.

H. Pipeline Overview

Fig. 1 summarizes the complete experimental pipeline from raw data to reported results.

TABLE I

Dataset characteristics after filtering. Base rates are the fraction of positive labels. The positive class is the adverse outcome for COMPAS and German Credit and the favourable outcome for Adult.

	COMPAS	German	Adult
Samples n	6,172	1,000	45,222
Encoded features d	12	21	79
Target ($y = 1$)	re-offend	bad credit	income > 50K
Sensitive attribute	race	sex	sex
Groups $ S $	6	2	2
Overall base rate	0.455	0.300	0.248
Two largest groups (size; base rate)			
Group 1	Afr.-Am. (3,175; 0.523)	male (690; 0.277)	Male (30,527; 0.312)
Group 2	Caucasian (2,103; 0.391)	female (310; 0.352)	Female (14,695; 0.114)

IV. Experimental Setup

A. Datasets

We use three datasets that together span three application domains, two sensitive attributes, and one and a half orders of magnitude in sample size. Table I reports their characteristics after filtering.

Base rates differ across groups in all three datasets, most sharply in Adult (0.312 versus 0.114, a factor of 2.7). This is the precise condition under which demographic parity and equalized odds are provably incompatible [8], and it drives the result of Section V-D.

1) COMPAS and a label-leakage correction: We apply ProPublica’s standard row filters [1] (screening date within ± 30 days of arrest, valid recidivism flag, charge degree not ordinary, valid score text) and use their standard feature set: age, sex, juvenile felony/misdemeanour/other counts, prior counts, charge degree, and race.

A feature set previously used in our own pipeline additionally contained duration = end – start, the length of the observation window. This feature correlates -0.78 with the target for a mechanical reason: re-offending truncates the observation window, so a short duration nearly implies the positive label. It is label leakage rather than signal, and we remove it. Doing so lowers COMPAS accuracy to approximately 0.68, consistent with the literature; we regard the lower number as the correct one and report it throughout.

2) German Credit: We take the Statlog encoding [25] and define $y = 1$ as bad credit risk, so that—as in COMPAS—the positive class is the adverse outcome.

3) Adult: We use the 48,842-row version [26], drop rows with missing values, exclude fnlwgt (a census sampling weight, not a person-level attribute), and prefer the ordinal education-num over the redundant categorical education.

The sensitive attribute is retained as a model input in all cases. This is a deliberate “fairness through awareness” choice [4]: removing it does not remove the information, since correlated proxies remain, and withholding it would prevent the group term (1) from being expressed.

B. Preprocessing and Protocol

Each seed $k \in \{0, \dots, 9\}$ draws an independent 60/20/20 train/validation/test split and initialises model weights. Numeric features are standardised and categorical features one-hot encoded with the first level dropped; unseen categories at transform time are encoded as all-zeros. All transformers are fit on the training split only. Splits are not stratified.

Ten seeds give ten paired observations per comparison. All results are reported as mean $\pm$ standard deviation over seeds.

C. Evaluation Metrics

We evaluate on hard test predictions at a fixed threshold of 0.5.

1) Accuracy: Fraction of correct predictions. Higher is better. There is no universal target; the relevant reference points are the majority-class rate and dataset-typical published performance.

2) Demographic parity difference (DPD): The largest gap in selection rate between groups, $\max_g P(\hat{y} = 1 \mid S = g) - \min_g P(\hat{y} = 1 \mid S = g)$. Lower is better; the range is $[0, 1]$. We emphasize DPD_2 , the same quantity computed on the two largest groups only. On COMPAS the all-groups statistic is pinned by groups of 31 and 11 individuals (roughly 6 and 2 in a test split), making it extremely unstable: across seeds we measure 0.608 ± 0.211 for the all-groups version against 0.304 ± 0.049 for DPD_2 , a fourfold reduction in variance. For German and Adult, which have two groups, the two statistics coincide by construction.

3) Equalized odds difference (EOD): The larger of the between-group true-positive-rate gap and the false-positive-rate gap [10]. Lower is better. Unlike DPD it conditions on the true label.

4) Generalized entropy index, $\text{GE}(2)$: Equation (3) evaluated on hard predictions, i.e. the inequality of the benefit vector (2) with $b_i \in \{0, 1, 2\}$. Lower is better, with 0 attained only by a perfect classifier.

We note explicitly that no threshold in the literature designates a “fair” value of DPD, EOD, or $\text{GE}(2)$. The four-fifths rule [27] is sometimes invoked, but it is a ratio test on selection rates and does not translate into a difference threshold. Values such as $\text{DPD} < 0.05$ are conventions of practice, not standards. $\text{GE}(2)$ has no absolute interpretation at all and is meaningful only in comparison between models on the same data. We therefore report all comparisons as differences against a matched baseline with significance tests, rather than against absolute cutoffs.

D. Baselines

We compare against one representative from each mitigation family plus two unconstrained accuracy references, all fit on the same per-seed split:

- Reweighting [12] (pre-processing): logistic regression trained with sample weights $w(s, y) = P(s)P(y)/P(s, y)$.
- ExponentiatedGradient [16] (in-processing), with demographic-parity and with equalized-odds constraints.
- ThresholdOptimizer [10] (post-processing), demographic-parity constraint, group-specific thresholds.
- Random Forest (300 trees) and XGBoost [28] (300 estimators, depth 6, learning rate 0.1): no fairness intervention; these establish the accuracy ceiling.

E. Statistical Testing

For each (fair model, baseline) pair we apply a paired Wilcoxon signed-rank test [29] across the ten seeds—paired because both models see identical splits, non-parametric because ten observations do not justify a normality assumption [30]. We correct across the four-metric family $\{\text{accuracy}, \text{DPD}_2, \text{EOD}, \text{GE}(2)\}$ using the Holm step-down procedure [31]. With $n = 10$ the smallest attainable two-sided p -value is $2^{-9} \approx 0.002$, which we report as $p = 0.002$ rather than as an open bound.

F. Implementation

Models and the composite loss are implemented in PyTorch [32]; preprocessing, logistic regression and random forests use scikit-learn [33]; group-fairness metrics and the ExponentiatedGradient and ThresholdOptimizer baselines use Fairlearn [34]; the generalized entropy index uses AI Fairness 360 [35]. The full grid is 7×7 values of (α, β) with $\alpha, \beta \in \{0, 0.005, 0.05, 0.25, 0.5, 0.75, 1\}$, over 2 architectures, 3 datasets and 10 seeds: 2,940 trained models. All runs are CPU-only.

V. Results

A. Surrogate Validation

Because training minimises the soft surrogates (1) and (3) while evaluation uses hard-threshold metrics, we first check that the two agree. Fig. 2 plots each surrogate against the metric it stands in for, over all 2,940 models.

Pearson correlations are $r = 0.794$ (COMPAS), 0.625 (German) and 0.907 (Adult) for $\mathcal{L}_{\text{DP}} \leftrightarrow \text{DPD}_2$, and 0.832, 0.919, 0.923 for $\mathcal{L}_{\text{GE}} \leftrightarrow \text{GE}(2)$. The surrogates track their targets, with German’s group surrogate the weakest link—an early indication of the small-sample behaviour analysed in Section V-F.

B. RQ1: A Double Dissociation Between the Two Objectives

The cleanest test of a trade-off holds the total fairness weight fixed and moves only β . Table III does this at the selected α^* , comparing the two extremes—individual-only ($\beta = 0$) and group-only ($\beta = 1$)—against the unconstrained $\alpha = 1$ baseline. Fig. 3 shows the same contrast as percentage change.

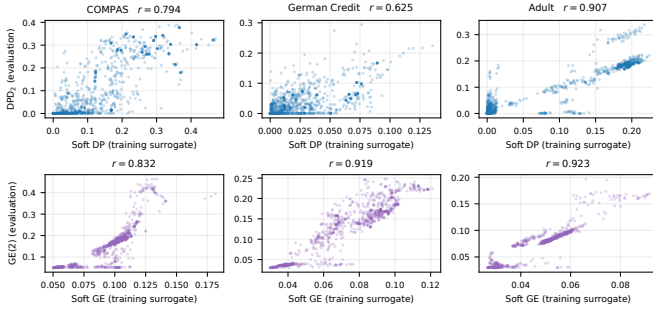


Fig. 2. Training surrogate versus hard evaluation metric, all sweep points and all seeds. Top: $\mathcal{L}_{DP}$ against DPD_2 . Bottom: $\mathcal{L}_{GE}$ against $GE(2)$. Correlations are strong on COMPAS and Adult; the weaker $r = 0.625$ for $\mathcal{L}_{DP}$ on German reflects that dataset’s 200-row test split rather than a defect of the surrogate.

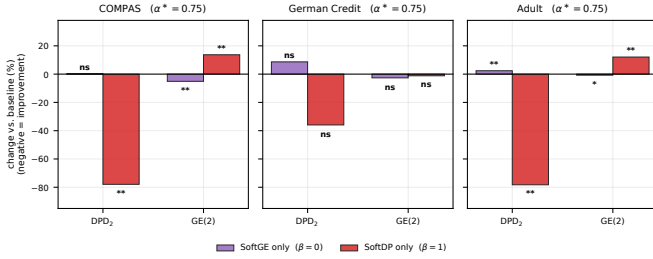


Fig. 3. Percentage change from the unconstrained baseline when the fairness budget is spent entirely on the individual term ($\beta = 0$, purple) or entirely on the group term ($\beta = 1$, red), logistic regression. The two objectives improve the metric they target and leave or push the other in the wrong direction. Markers give Holm-corrected paired Wilcoxon significance (** $p < 0.01$, * $p < 0.05$, ns = not significant).

The pattern is a double dissociation and it is consistent on the two datasets with a disparity to act on:

- Group objective ($\beta = 1$) reduces DPD_2 by 78% on COMPAS ($0.304 \rightarrow 0.067$) and 78% on Adult ($0.184 \rightarrow 0.040$), both $p_{Holm} = 0.008$. It simultaneously worsens $GE(2)$ ($0.191 \rightarrow 0.217$ on COMPAS, $0.084 \rightarrow 0.094$ on Adult), both $p_{Holm} = 0.008$ —i.e. worse individual fairness than doing nothing at all.
- Individual objective ($\beta = 0$) improves $GE(2)$ ($0.191 \rightarrow 0.181$ on COMPAS, $0.084 \rightarrow 0.083$ on Adult, both $p_{Holm} \leq 0.012$) but does not improve DPD_2 : it is statistically unchanged on COMPAS ($0.304 \rightarrow 0.305$, $p = 0.85$) and significantly worse on Adult ($0.184 \rightarrow 0.189$, $p_{Holm} = 0.008$).

The effects are asymmetric in size: the group term moves its target by nearly 80%, the individual term by around 5%. We interpret this in Section VI.

C. RQ2: Consistency Across Datasets and Architectures

Fig. 4 traces both metrics continuously across β at fixed $\alpha = 0.75$. On COMPAS and Adult, and for both architectures, the two curves cross: as β increases the group metric falls monotonically while the individual metric rises monotonically. The MLP reproduces the

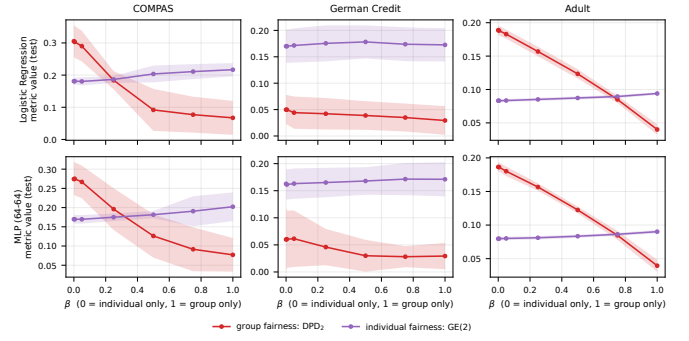


Fig. 4. Group fairness (DPD_2 , red) and individual fairness ($GE(2)$, purple) as β moves from individual-only (0) to group-only (1), at $\alpha = 0.75$. Bands are ± 1 standard deviation over ten seeds. The crossing curves on COMPAS and Adult are the trade-off; German Credit is flat because its baseline disparity is already small.

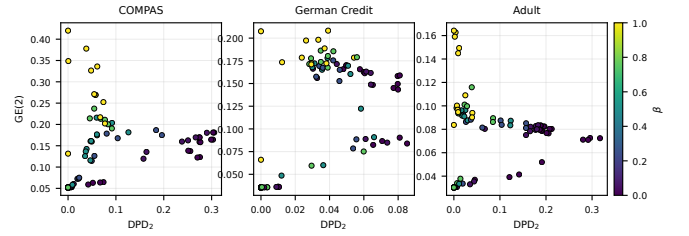


Fig. 5. Group versus individual fairness across all $\alpha < 1$ sweep points (seed means), coloured by β . Points order along the diagonal by β , tracing the attainable trade-off surface.

logistic regression pattern closely, indicating the tension is not an artefact of a linear hypothesis class.

Fig. 5 views the same data as a frontier in (DPD_2 , $GE(2)$) space, with β encoded by colour: configurations sort along a diagonal from low- $GE(2)$ /high- DPD_2 to high- $GE(2)$ /low- DPD_2 . Appendix heatmaps (Figs. 9–10) show the full $\alpha \times \beta$ surface for both metrics.

D. The Adult Parity-Odds Inversion

Adult exhibits the trade-off in its sharpest form, involving two group criteria. Driving demographic parity down drives equalized odds up: for the MLP at $\beta = 1$, DPD_2 falls from 0.182 to 0.009—essentially perfect demographic parity—while EOD rises from 0.097 to 0.321, a factor of 3.3 (both $p_{Holm} = 0.008$). For logistic regression the same contrast gives $0.184 \rightarrow 0.040$ on DPD_2 against $0.107 \rightarrow 0.246$ on EOD.

This is the empirical face of a known impossibility. Adult’s group base rates differ by a factor of 2.7 (Table I), and under unequal base rates independence and separation cannot hold together [8]. The contribution here is the exchange rate: on this dataset roughly one point of demographic parity is bought with 1.1 points of equalized odds. A model can therefore be made near-perfectly “fair” under one standard definition while becoming markedly less fair under another, and the reported number depends

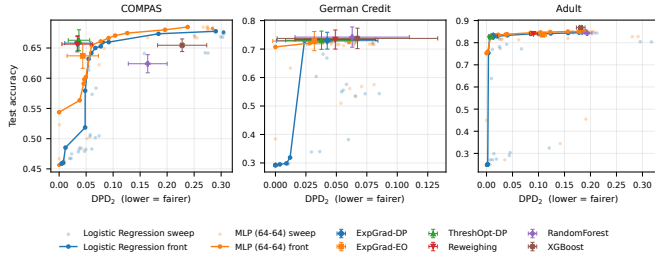


Fig. 6. Accuracy versus DPD_2 (seed means). Faint points are individual sweep configurations; solid lines are the Pareto fronts per architecture; markers with error bars are the baselines (± 1 std over seeds). The composite-loss front reaches the same region as the dedicated group-fairness baselines.

entirely on which definition the deploying institution has adopted.

E. RQ3: Comparison Against Established Baselines

Table II reports every method on every dataset, and Fig. 6 places the full sweep and the baselines in accuracy- DPD_2 space.

The honest reading is that the composite loss is competitive but not superior on group fairness. On COMPAS at comparable accuracy, Reweighting reaches $DPD_2 = 0.034$ and ExponentiatedGradient 0.035, against 0.061 for our selected model; on Adult, ThresholdOptimizer reaches 0.009 and ExponentiatedGradient 0.013, against 0.021. Our MLP at (0.836, 0.026) is nonetheless on the joint frontier, being more accurate than every baseline that achieves lower disparity.

Two further observations are worth recording. First, the accuracy ceiling on Adult belongs to XGBoost at 0.868, which also carries one of the largest disparities ($DPD_2 = 0.181$)—the most accurate model is among the least fair. Second, ExponentiatedGradient with an equalized-odds constraint behaves on Adult exactly as the inversion of Section V-D predicts, in mirror image: it attains the best EOD of any method (0.035) while leaving DPD_2 at 0.108, an order of magnitude above the parity-constrained methods.

F. RQ2 Continued: The German Credit Null Result and Its Cause

On German Credit, no comparison between a selected fair model and its baseline survives Holm correction on any metric for either architecture (all $p_{\text{Holm}} \geq 0.42$; see Fig. 7). We report this as a negative result and diagnose it, because the cause is not that the method fails.

Two mechanisms combine, and Table IV quantifies both.

1) Validation demographic parity is uninformative at $n = 1000$: Selection ranks configurations by validation DPD_2 . On German that quantity is essentially uncorrelated with the test value it is meant to predict ($r = 0.08$ for logistic regression, 0.11 for the MLP), against $r \geq 0.89$ on COMPAS and $r \geq 0.99$ on Adult (Fig. 8). With a

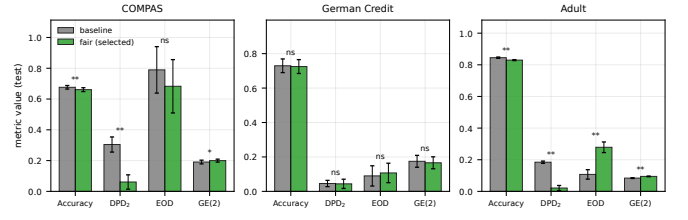


Fig. 7. Selected fair logistic regression against its unconstrained baseline (mean $\pm$ std over ten seeds), with Holm-corrected paired Wilcoxon markers. COMPAS and Adult show significant reductions in DPD_2 ; German shows no significant change on any metric.

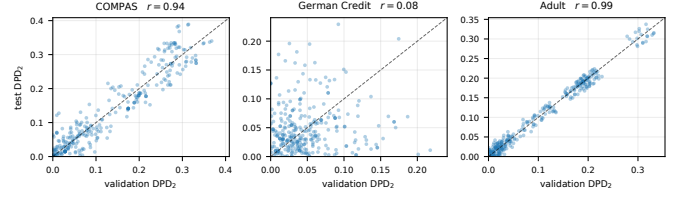


Fig. 8. Validation versus test DPD_2 across all non-degenerate sweep configurations (logistic regression). Dashed line is $y = x$. Validation demographic parity predicts its test counterpart on COMPAS and Adult but carries almost no information on German Credit.

200-row validation split containing roughly 62 women, the standard error of a group selection rate is about ± 0.04 —the size of the entire signal. The between-configuration spread is smaller than the within-configuration seed noise (signal-to-noise 0.45 versus 1.68 and 12.9). Selection is therefore ranking noise.

2) The selection criterion is scale-sensitive: The utopia distance $\sqrt{(1 - \text{acc})^2 + DPD_2^2}$ weights an accuracy point and a disparity point equally. That is reasonable when the two terms are comparable, as on COMPAS (1.1 $\times$) and Adult (0.7 $\times$), but on German the accuracy term exceeds the disparity term by a factor of 34.9. The criterion degenerates to “maximise accuracy” and is nearly blind to fairness even before noise is considered. Consistent with this, the selected β averages only 0.19 on German—with three of ten seeds selecting $\beta = 0$, i.e. no group-fairness weight at all—against 0.63 on COMPAS and 0.58 on Adult, and the selected configuration differs across 8 of 10 seeds.

3) The effect exists under a pre-specified configuration: Bypassing selection entirely and fixing the group-fairness extreme $\beta = 1$, German logistic regression attains $DPD_2 = 0.024$ against a baseline 0.046—a 48% reduction—at no accuracy cost (0.729 versus 0.730), improving in 8 of 10 seeds ($p = 0.020$, uncorrected). The pre-specified ablation contrast in Table III shows the same direction (0.046 $\rightarrow$ 0.029) though it does not survive correction. We conclude that the method reduces disparity on German, and that validation-based selection at this sample size cannot find the configurations that do so.

TABLE II

Master comparison. Mean $\pm$ standard deviation over ten seeds, written $x \pm \sigma$. Arrows give the preferred direction. DPD₂ is the demographic parity difference on the two largest groups; for German and Adult it coincides with the all-groups value. Best value per column among all methods is in bold. Baseline families: Reweighting is pre-processing, ExpGrad in-processing, ThreshOpt post-processing. Fair models select (α, β) on validation data only.

Method	COMPAS (race)				German Credit (sex)				Adult (sex)			
	Acc $\uparrow$	DPD ₂ $\downarrow$	EOD $\downarrow$	GE(2) $\downarrow$	Acc $\uparrow$	DPD ₂ $\downarrow$	EOD $\downarrow$	GE(2) $\downarrow$	Acc $\uparrow$	DPD ₂ $\downarrow$	EOD $\downarrow$	GE(2) $\downarrow$
Unconstrained baselines												
Logistic regression	.676 $\pm$.012	.304 $\pm$.049	.789 $\pm$.151	.191 $\pm$.012	.730 $\pm$.040	.046 $\pm$.018	.090 $\pm$.059	.175 $\pm$.035	.845 $\pm$.004	.184 $\pm$.007	.107 $\pm$.030	.084 $\pm$.002
MLP (64-64)	.682 $\pm$.014	.284 $\pm$.044	.829 $\pm$.151	.177 $\pm$.011	.724 $\pm$.034	.060 $\pm$.052	.106 $\pm$.064	.164 $\pm$.027	.851 $\pm$.004	.182 $\pm$.008	.097 $\pm$.024	.080 $\pm$.003
Random Forest	.624 $\pm$.015	.164 $\pm$.036	.808 $\pm$.205	.195 $\pm$.011	.742 $\pm$.036	.063 $\pm$.047	.133 $\pm$.086	.156 $\pm$.027	.843 $\pm$.002	.193 $\pm$.009	.104 $\pm$.021	.083 $\pm$.001
XGBoost	.655 $\pm$.010	.228 $\pm$.045	.791 $\pm$.181	.196 $\pm$.011	.738 $\pm$.036	.067 $\pm$.066	.152 $\pm$.102	.148 $\pm$.024	.868 $\pm$.003	.181 $\pm$.008	.087 $\pm$.023	.070 $\pm$.002
This work: composite loss												
Fair LogReg (ours)	.661 $\pm$.013	.061 $\pm$.047	.683 $\pm$.173	.200 $\pm$.010	.725 $\pm$.040	.044 $\pm$.027	.107 $\pm$.057	.166 $\pm$.035	.830 $\pm$.003	.021 $\pm$.016	.279 $\pm$.034	.095 $\pm$.002
Fair MLP (ours)	.669 $\pm$.018	.082 $\pm$.047	.766 $\pm$.181	.181 $\pm$.014	.716 $\pm$.035	.069 $\pm$.052	.127 $\pm$.082	.162 $\pm$.036	.836 $\pm$.004	.026 $\pm$.015	.281 $\pm$.031	.091 $\pm$.003
Established mitigation baselines												
Reweighting	.656 $\pm$.013	.034 $\pm$.029	.644 $\pm$.137	.205 $\pm$.011	.734 $\pm$.034	.049 $\pm$.033	.118 $\pm$.078	.162 $\pm$.025	.840 $\pm$.003	.090 $\pm$.007	.149 $\pm$.029	.088 $\pm$.002
ExpGrad (DP)	.658 $\pm$.012	.035 $\pm$.024	.766 $\pm$.216	.200 $\pm$.010	.730 $\pm$.030	.042 $\pm$.042	.104 $\pm$.086	.163 $\pm$.022	.830 $\pm$.003	.013 $\pm$.008	.300 $\pm$.014	.096 $\pm$.002
ExpGrad (EO)	.637 $\pm$.021	.044 $\pm$.029	.737 $\pm$.206	.206 $\pm$.015	.728 $\pm$.034	.032 $\pm$.034	.106 $\pm$.054	.164 $\pm$.027	.836 $\pm$.002	.108 $\pm$.007	.035 $\pm$.015	.090 $\pm$.003
ThreshOpt (DP)	.663 $\pm$.017	.037 $\pm$.020	.781 $\pm$.208	.181 $\pm$.010	.731 $\pm$.032	.038 $\pm$.029	.129 $\pm$.060	.162 $\pm$.025	.827 $\pm$.003	.009 $\pm$.006	.333 $\pm$.021	.097 $\pm$.003

TABLE III

Ablation over β at the selected α^* : the fairness budget is held fixed and redistributed between the two notions. Mean $\pm$ std over ten seeds. Significance is a paired Wilcoxon test against the $\alpha = 1$ row of the same block, Holm-corrected across the four metrics (** $p < 0.01$, * $p < 0.05$; unmarked = not significant). The double dissociation is visible in the DPD₂ and GE(2) columns: $\beta = 1$ improves the former and worsens the latter, $\beta = 0$ does the reverse.

Dataset	Variant	Logistic Regression				MLP (64-64)			
		Acc $\uparrow$	DPD ₂ $\downarrow$	EOD $\downarrow$	GE(2) $\downarrow$	Acc $\uparrow$	DPD ₂ $\downarrow$	EOD $\downarrow$	GE(2) $\downarrow$
COMPAS	No fairness ($\alpha=1$)	.676 $\pm$.012	.304 $\pm$.049	.789 $\pm$.151	.191 $\pm$.012	.682 $\pm$.014	.284 $\pm$.044	.829 $\pm$.151	.177 $\pm$.011
	SoftGE only ($\beta=0$)	.674 $\pm$.012	.305 $\pm$.049	.811 $\pm$.135*	.181 $\pm$.010**	.685 $\pm$.014	.274 $\pm$.041	.842 $\pm$.138	.170 $\pm$.011**
	SoftDP only ($\beta=1$)	.650 $\pm$.014**	.067 $\pm$.053**	.681 $\pm$.167	.217 $\pm$.021**	.661 $\pm$.023**	.077 $\pm$.044**	.634 $\pm$.171*	.202 $\pm$.037**
	Blend ($\beta=\beta^*$)	.660 $\pm$.021**	.092 $\pm$.065**	.678 $\pm$.159	.204 $\pm$.026*	.666 $\pm$.023*	.092 $\pm$.057**	.632 $\pm$.186*	.191 $\pm$.039
German	No fairness ($\alpha=1$)	.730 $\pm$.040	.046 $\pm$.018	.090 $\pm$.059	.175 $\pm$.035	.724 $\pm$.034	.060 $\pm$.052	.106 $\pm$.064	.164 $\pm$.027
	SoftGE only ($\beta=0$)	.730 $\pm$.037	.050 $\pm$.028	.106 $\pm$.066	.170 $\pm$.032	.716 $\pm$.033	.080 $\pm$.071	.146 $\pm$.067	.158 $\pm$.029
	SoftDP only ($\beta=1$)	.731 $\pm$.036	.029 $\pm$.027	.103 $\pm$.070	.173 $\pm$.031	.726 $\pm$.032	.038 $\pm$.032	.124 $\pm$.102	.172 $\pm$.028
	Blend ($\beta=\beta^*$)	.728 $\pm$.038	.044 $\pm$.031	.105 $\pm$.070	.171 $\pm$.032	.721 $\pm$.035	.039 $\pm$.043	.105 $\pm$.056	.168 $\pm$.030
Adult	No fairness ($\alpha=1$)	.845 $\pm$.004	.184 $\pm$.007	.107 $\pm$.030*	.084 $\pm$.002	.851 $\pm$.004	.182 $\pm$.008	.097 $\pm$.024	.080 $\pm$.003
	SoftGE only ($\beta=0$)	.845 $\pm$.004	.189 $\pm$.007**	.113 $\pm$.030*	.083 $\pm$.002*	.851 $\pm$.004	.191 $\pm$.007**	.097 $\pm$.015	.079 $\pm$.002**
	SoftDP only ($\beta=1$)	.833 $\pm$.003**	.040 $\pm$.008**	.246 $\pm$.023**	.094 $\pm$.002**	.831 $\pm$.004**	.009 $\pm$.005**	.321 $\pm$.023**	.095 $\pm$.003**
	Blend ($\beta=\beta^*$)	.833 $\pm$.003**	.040 $\pm$.008**	.246 $\pm$.023**	.094 $\pm$.002**	.836 $\pm$.004**	.023 $\pm$.012**	.290 $\pm$.030**	.091 $\pm$.003**

Selected α^* : COMPAS 0.75/0.75, German 0.75/0.5, Adult 0.75/0.5 (LR/MLP). Selected β^* : COMPAS 0.5/0.75, German 0.05/0.75, Adult 1.0/0.5. For Adult logistic regression $\beta^* = 1$, so the blend row coincides with the group-only row.

TABLE IV

Why selection fails on German Credit. r is the correlation between validation and test DPD₂ over non-degenerate sweep configurations; SNR is the between-configuration spread of validation DPD₂ divided by its within-configuration seed noise; the last column is the ratio of the accuracy term to the disparity term in the utopia-distance criterion.

Dataset	Arch.	$r_{\text{val}, \text{test}}$	SNR	acc:DPD term
COMPAS	LR	0.94	1.68	1.1 $\times$
	MLP	0.89	1.54	1.3 $\times$
German	LR	0.08	0.45	34.9$\times$
	MLP	0.11	0.51	21.0$\times$
Adult	LR	0.99	12.91	0.7 $\times$
	MLP	0.99	7.19	0.7 $\times$

G. Supplementary: Sweep Surfaces

Figs. 9 and 10 give the full seed-averaged $\alpha \times \beta$ surfaces. The $\alpha = 0$ row is retained for completeness but is excluded

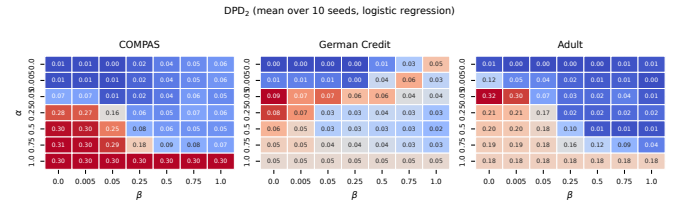


Fig. 9. DPD₂ over the $\alpha \times \beta$ grid (mean over ten seeds, logistic regression). Disparity falls as β increases, most strongly at intermediate α .

from selection: with no accuracy term the optimiser produces near-constant predictors whose apparent fairness is degenerate.

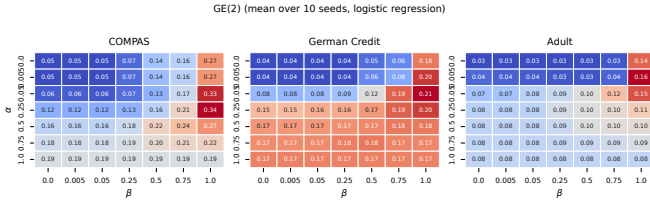


Fig. 10. $GE(2)$ over the same grid. The gradient runs in the opposite direction to Fig. 9 across β , which is the double dissociation viewed as a surface.

VI. Discussion

A. Why the Two Objectives Are Not Symmetric

The group term moves its target by roughly 80%; the individual term moves its own by roughly 5%. The asymmetry is not a tuning artefact but follows from what each objective can control.

$\mathcal{L}_{DP}$ in (1) is a difference of group means of p . A model can reduce it by shifting the decision surface between groups—an operation that costs little accuracy because it redistributes predictions rather than requiring better ones. The quantity is, in effect, freely adjustable.

$\mathcal{L}_{GE}$ in (3) is the inequality of $b_i = 1 + p_i - y_i$. Benefit equality is attained exactly when b_i is constant, that is when every individual is predicted equally correctly. Its floor is therefore set by the irreducible error of the hypothesis class on the data: one cannot equalize benefit without in fact being more accurate. On COMPAS, where the ceiling is approximately 0.68 accuracy, there is little room, and the term accordingly achieves little. A useful implication is that individual fairness in this outcome-based sense is not primarily an allocation problem but a prediction-quality problem.

B. β as a Policy Parameter, Not a Hyperparameter

Hyperparameters are tuned: one searches for the value that maximises a criterion. β does not admit that treatment, because the two ends of its range optimise different criteria and there is no metric that ranks them. Selecting β by validation performance simply encodes whichever fairness notion the validation metric happens to measure—which is exactly what our utopia-distance rule does, and why it is scale-sensitive (Section V-F).

β is better understood as a policy dial: it makes explicit a commitment that is otherwise implicit and undocumented in a deployed system. Our results give that commitment a price. On Adult, moving from $\beta = 0$ to $\beta = 1$ buys a 0.18 reduction in demographic parity difference at a cost of 0.22 in equalized odds difference and 0.015 in benefit inequality. An institution that must state which definition it is optimizing can read the consequences of that choice off the trade-off map rather than discovering them after deployment.

C. The Uncomfortable Case

The Adult MLP at $\beta = 1$ makes the tension concrete. It achieves $DPD_2 = 0.009$: men and women are selected at nearly identical rates, and by the most widely cited group-fairness definition the model is close to ideal. The same model has $EOD = 0.321$, meaning its error rates differ across sexes by roughly a third—worse than the unconstrained baseline by a factor of 3.3. Both statements are true of the same predictor.

There is no analysis internal to the data that resolves this. Whether it is better to select equally or to err equally is a normative question about what the decision is for. The value of a trade-off map is not that it answers the question but that it prevents the answer from being made accidentally, by a default in a fairness library.

D. Where the Composite Loss Sits Among Existing Methods

On group-fairness metrics the loss matches rather than beats Reweighting [12], ExponentiatedGradient [16] and ThresholdOptimizer [10] (Table II, Fig. 6). We regard that as sufficient for its purpose: an instrument used to measure a trade-off must be credible on the axis it shares with existing tools, and it is.

Its distinguishing property is the one those methods lack. Each of the baselines is constructed around a single constraint—demographic parity or equalized odds—and offers no continuous path to an individual-fairness objective. The composite loss traverses that path within a fixed architecture, optimizer and evaluation protocol, so differences along β are attributable to the objective rather than to a change of method. That is what makes Fig. 4 interpretable as a trade-off curve.

E. A Sample-Size Floor for Fairness Model Selection

The German Credit analysis (Section V-F) generalizes beyond this paper. Fairness metrics are differences of conditional rates, so their effective sample size is governed by the smallest group, not by n . With 310 women in the dataset and roughly 62 in a validation split, the standard error of a group selection rate is comparable to the entire disparity being measured, and validation-based tuning ranks noise (Table IV).

Practitioners frequently select fairness hyperparameters exactly this way. Our results suggest that such a procedure should be accompanied by a check—the validation-test correlation of the fairness metric across configurations is a cheap one—and that where it fails, a pre-specified configuration is preferable to a tuned one. The cost of ignoring this is not merely a weaker result: on German the tuned model reproduced the baseline disparity while a pre-specified $\beta = 1$ reduced it by 48% at no accuracy cost.

VII. Limitations

We state the constraints on these results explicitly, since several bear directly on how the numbers should be read.

1) The individual-fairness notion is outcome-based, not Lipschitz: We measure individual fairness as inequality of the benefit vector following Speicher et al. [5]. This requires no similarity metric, which is why it is computable here, but it does not establish the “similar individuals treated similarly” property of Dwork et al. [4]. Claims in this paper about individual fairness should be read as claims about the distribution of benefit.

2) GE(2) on hard predictions is a function of the error profile: With $\hat{y} \in \{0, 1\}$ the benefit takes only three values, $b_i \in \{0, 1, 2\}$, so GE(2) is determined by the counts of false positives and false negatives. It is a legitimate inequality measure of the error distribution, but it does not distinguish which individuals bear those errors.

3) Equalized odds on COMPAS is unreliable: Unlike DPD_2 , our EOD is computed over all six race groups, two of which contain 31 and 11 individuals in total. The resulting statistic has a standard deviation of 0.13–0.22 across seeds and no COMPAS EOD comparison reaches significance. COMPAS EOD values in Tables II and III should be treated as uninformative; a largest-two-groups EOD would be the appropriate fix and is left to future work.

4) German Credit is underpowered: At $n = 1000$ the test split contains 200 rows and validation-based selection does not function (Section V-F). The dataset supports the pre-specified contrast we report but not tuned model selection.

5) Full-batch training: Because (1) requires every group to be present to form a group gap, each epoch is one full-batch gradient step, capped at 300. Mini-batch behaviour of the composite loss—including whether stochastic group composition destabilises $\mathcal{L}_{\text{DP}}$ —is untested and would be required before applying this at larger scale.

6) Fixed threshold: The differentiable models decide at a fixed 0.5 threshold, whereas the ThresholdOptimizer baseline is permitted group-specific thresholds. The comparison in Table II is therefore not threshold-matched.

7) The selection rule embeds a value judgement: Minimising $\sqrt{(1 - \text{acc})^2 + \text{DPD}_2^2}$ treats one point of accuracy as equal to one point of disparity, and is sensitive to the scale of each term (Table IV). A different scalarization would select different models. The ablation results of Section V-B, which fix β rather than selecting it, do not depend on this choice.

8) Single, binary sensitive attribute: Each dataset is analysed with respect to one attribute, and no intersectional analysis (for example race $\times$ sex on Adult) is performed. Subgroup fairness is known to behave differently from marginal group fairness [36].

9) Inconsistent positive-class polarity: The positive class is the adverse outcome on COMPAS and German Credit and the favourable outcome on Adult. Directions of the fairness metrics are unaffected, but a statement such as “group A is selected more often” has opposite welfare meanings across datasets.

10) Sensitive attribute used as an input: Retaining the sensitive attribute is deliberate and standard in this setting, but it is a modelling choice with legal implications in some jurisdictions, and results may differ where its use is prohibited.

11) Adversarial debiasing not evaluated: All three mitigation families are represented, but a second in-processing method [17] was not run.

12) Scope: Conclusions are limited to binary classification on tabular data with the three datasets studied. Benchmark-specific artefacts are documented [21]–[23], and the COMPAS leakage correction of Section IV is one such case we encountered directly.

VIII. Conclusion

We set out to measure, rather than derive, the tension between group and individual fairness. To do so we built a differentiable composite loss whose parameter β interpolates continuously between a soft demographic-parity term and a soft generalized-entropy term, and swept it across a 7×7 (α, β) grid, three datasets, two model families and ten seeds: 2,940 trained models with paired significance testing throughout.

Three findings stand out. First, the two objectives exhibit a statistically significant double dissociation: the group term reduces demographic disparity by 78%–89% while significantly worsening benefit inequality, and the individual term significantly improves benefit inequality while leaving disparity unchanged or worse. The effects are strongly asymmetric in magnitude, because equalizing benefit is bounded by the achievable prediction quality in a way that equalizing selection rates is not. Second, on Adult the conflict extends within group fairness: driving demographic parity to 0.009 raises equalized odds by a factor of 3.3, an empirical instance of a known impossibility, with the exchange rate quantified. Third, where a baseline already exhibits little disparity, the intervention has nothing to act on, and where the dataset is small, validation-based selection cannot reliably find the configurations that would act—on German Credit the tuned model reproduced the baseline disparity while a pre-specified configuration reduced it by 48% at no accuracy cost.

The composite loss reaches group-fairness levels comparable to established pre-, in- and post-processing methods without surpassing them. That was not the objective; the objective was a measurement instrument credible on the axis it shares with existing tools, and one that additionally traverses the axis they do not.

The practical consequence is that β should not be tuned. There is no metric that ranks its endpoints, because they optimize different and partly opposed definitions of the same word. It is a policy parameter, and the contribution of a trade-off map is to attach a measured price to each setting so that the choice is made deliberately rather than inherited from a library default.

Future work should extend the map to intersectional subgroups, replace the all-groups equalized odds statistic with a largest-two-groups variant on COMPAS, test the composite loss under mini-batch training at larger scale, and characterise more precisely the sample-size floor below which fairness model selection ceases to function.

Reproducibility

All experiments run on CPU. The complete pipeline—data loading, the $\alpha \times \beta$ sweep, baselines, significance testing, and every figure and table in this paper—is reproducible with two commands, and per-seed result files are written for all 2,940 trained models.

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